

Mohammed Nihad

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EDUCATION

University of Virginia (committed transfer, Fall 2026)

August 2026 - May 2028

- Bachelor of Science in Computer Engineering

City College of New York

August 2024 - May 2026

- Bachelor of Science in Computer Engineering (5x Hackathon winner, 3x dean's list)
- Relevant Coursework:** Distributed Systems, Data Structures and Algorithms, Software Engineering and Design, Database Systems

Skills

- Languages:** Python, C++, TypeScript, Java, MATLAB, SQL
- Frameworks:** FastAPI, Spring Boot, React.js, React Native, PyTorch, scikit-learn, Next.js
- Tools:** Git/ GitHub, AWS, Docker, Figma, PostgreSQL, MongoDB, LangChain, Kubernetes, Grafana

EXPERIENCE

Bloom Energy, Software Engineering Intern

June 2026 – August 2026

- Unified millions of records across Salesforce, OSI PI, AWS Redshift, and internal systems into an internal diagnostic tool, enabling RMCC engineers to look up active escalations, match historical fault patterns, and receive root cause analysis with fix recommendations, cutting troubleshooting time by 80% across 30,000 monitored fuel cells.
- Built and shipped a production visual workflow builder using React Flow, FastAPI, and MongoDB, enabling engineers to assemble reusable node-based DAGs and reducing hand-authoring time for new automation scripts by ~80%.
- Architected a versioned compiler pipeline that translated DAGs into validated, Ruff-linted notebooks, configs, and pipeline artifacts, incorporating cycle detection and lazy-cached introspection; adopted across a team of 12.
- Optimized AWS Redshift queries by refactoring nested query patterns and simplifying data-access logic, reducing query runtime from ~50s to <10s; then built an MCP server to expose the data to engineers in local dev environments.

Rehost, Software Engineering Intern

December 2025 – March 2026

- Improved 30-day returning users by building an agentic Retention Autopilot that ingests analytics, segments users, and launches approved A/B-tested push + rewards campaigns, cutting campaign setup time by 30%.

STEMKasa, Software Engineering Intern

June 2025 – August 2025

- Led 3 interns to rebuild a 17-file monolith as a MERN app, implementing component-based design, REST APIs, and integration tests to preserve 100% functionality.
- Architected and deployed a full CI/CD pipeline using GitHub Actions, automating build, test, and deployment workflows to eliminate manual deployment errors across staging and production environments.
- Engineered an AI tutoring layer with GPT-powered adaptive flashcards and lessons, then integrated secure auth and Stripe subscriptions, driving a 25% engagement increase across 3 schools.

Research Paper

June 2025 – December 2025

- Co-authored a Research paper on a self-critiquing LLM pipeline (expert feedback + distillation + RL, Pareto optimization) in PyTorch that improved accuracy while cutting infra costs up to 90% with open-weight models.

Universacare, Software Automation Intern

June 2023 – September 2023

- Built a Python Selenium web scraping workflow to extract daily Home Health Aide data, automating manual work and improving productivity by 11%.
- Developed 15 reusable React components and integrated a role-aware online application form into the production website, improving UI consistency/access control and saving 2+ hours per submission.

Projects

Squeeze (Morgan Stanley 2026 Hackathon winner)

FastAPI | LangChain | LangGraph | Next.js | Supabase

- Built a scalable event-management platform with FastAPI/Next.js backend services, tool orchestration, and AWS Bedrock-powered automation across 12 integrated workflows.
- Platform acquired by Lemontree (Hackathon partners), now serving 500K+ users in production

RhetoriQ ([Source Code](#))

Kafka | Flink | Kubernetes | Neo4j | pgvector | LangChain

- Built a low-latency narrative-analysis pipeline using Kafka/Flink, Neo4j graph traversal, pgvector semantic search, Redis caching, and Kubernetes microservices to process 100K+ daily documents.
- Designed a multi-agent orchestration system of 7 specialized AI agents, spanning planning, search, investigation, and synthesis that autonomously trace the origin and spread of political.
- Developed an agentic LangChain investigation pipeline leveraging HuggingFace NLP models for semantic similarity detection, with Prometheus/Grafana observability for 15+ microservices.